Speech LLM architectures, and an evaluated Indic speech → text-response pipeline

Technical assignment, Speech AI Research Internship, Vaani Research Lab. Code and reproduction steps: README in the repository.

Task 1 — Comparative architecture analysis

> Models: **Moshi** (Kyutai, 2024; full-duplex speech-to-speech) and **Qwen3-Omni-30B-A3B** (Alibaba, 2025; turn-based Thinker–Talker speech-to-speech). Both generate speech. All figures below are quoted from the two technical reports (arXiv 2410.00037, 2509.17765) unless marked *inference*. Numbers for latency are the authors' own, measured on their hardware; they are not comparable with my laptop measurements in Task 2.

1.1 Side-by-side

	Moshi	**Qwen3-Omni**
Speech encoder	Mimi, a causal streaming codec: conv encoder + Transformer, 24 kHz in, 12.5 Hz frames (80 ms), 8-level RVQ, 1.1 kbps. Encoder and decoder both streaming.	AuT (Audio Transformer), ~0.6B attention encoder-decoder trained *from scratch* on 20M h of supervised audio (80% zh/en pseudo-labelled ASR, 10% other-language ASR, 10% audio understanding); 128-ch mel, 8× Conv2D downsampling to 12.5 Hz; dynamic attention window of 1–8 s.
Semantic vs acoustic	Explicitly split inside the codec: RVQ level 1 is distilled from WavLM (semantic); levels 2–8 carry acoustic detail. A text “inner monologue” adds a third, linguistic channel.	Input side is a *supervised, ASR-trained* representation, so it is largely semantic. Output side is a multi-codebook RVQ codec (codebook 0 from the Talker, residuals from an MTP module). The report does not describe a separate semantic/acoustic split on the output codec.
Speech→LLM alignment	No adapter or projector. The Temporal Transformer takes, at every 80 ms step, the *sum* of learned embeddings of all audio sub-streams (user and Moshi) plus the text token. Alignment is learned by joint training from a text-pretrained 7B (Helium, 2.1T tokens).	Encoder features enter the Thinker (MoE 30B-A3B) as a token sequence, positioned by TM-RoPE with absolute 80 ms time IDs shared across audio and video. Training: S1 encoder alignment (LLM frozen), S2 general, S3 long-context (32k).
Streaming	Fully streaming in and out, frame by frame; listens while speaking (two parallel streams, 3000-step / ~4 min context).	Streaming via chunked prefill: Thinker and Talker prefill chunks asynchronously while the user is still talking; output streams from the first codec frame. Turn-based: one stream in, then one reply.
AR vs NAR generation	Autoregressive everywhere. Temporal Transformer steps over time; a small Depth Transformer (6 layers, d=1024) steps over the 8 codebooks of one frame. Acoustic delay of 1 frame (2 in pre-training) lets semantic tokens lead.	Autoregressive Talker (3B-A0.3B) predicts codebook 0 per frame; an ~80M MTP module fills residual codebooks (fixed-step AR); Code2Wav is a ~200M *causal ConvNet* (non-AR) that replaced block-wise diffusion to enable frame-level streaming.
End-of-speech / stop	No end-of-turn token. The model always emits audio; “not speaking” is near-silent waveform with PAD in the text stream. Starting to talk = sampling EPAD. Turn-taking, overlap and interruption are learned from Fisher two-channel data.	The report does not describe user end-of-speech detection or barge-in. *Inference:* the Thinker–Talker split, with external modules allowed to intervene on text, suggests endpointing (e.g. VAD) lives outside the model; the model stops by emitting an end token. Treat this as unverified.

Objectives and data	Next-token cross-entropy on text, semantic and acoustic streams (text loss down-weighted). 7M h of mostly-English audio, transcribed with Whisper large-v3; 2,000 h Fisher (two-channel); 170 h supervised multi-stream; >20k h synthetic instruct dialogue made with a multi-stream TTS.	Multi-stage next-token training; Talker has four stages (hundreds of millions of speech samples → continual pre-training on high-quality data to reduce hallucination → DPO → speaker fine-tuning). Speech understanding in 19 languages, generation in 10.
Reported latency	160 ms theoretical (80 ms frame + 80 ms acoustic delay), ~200 ms practical.	234 ms (audio) / 547 ms (video) first packet at 1 stream; 728 / 1517 ms at 4 streams; 1172 / 2284 ms at 6. Generation RTF 0.47–0.66.

1.2 Diagrams

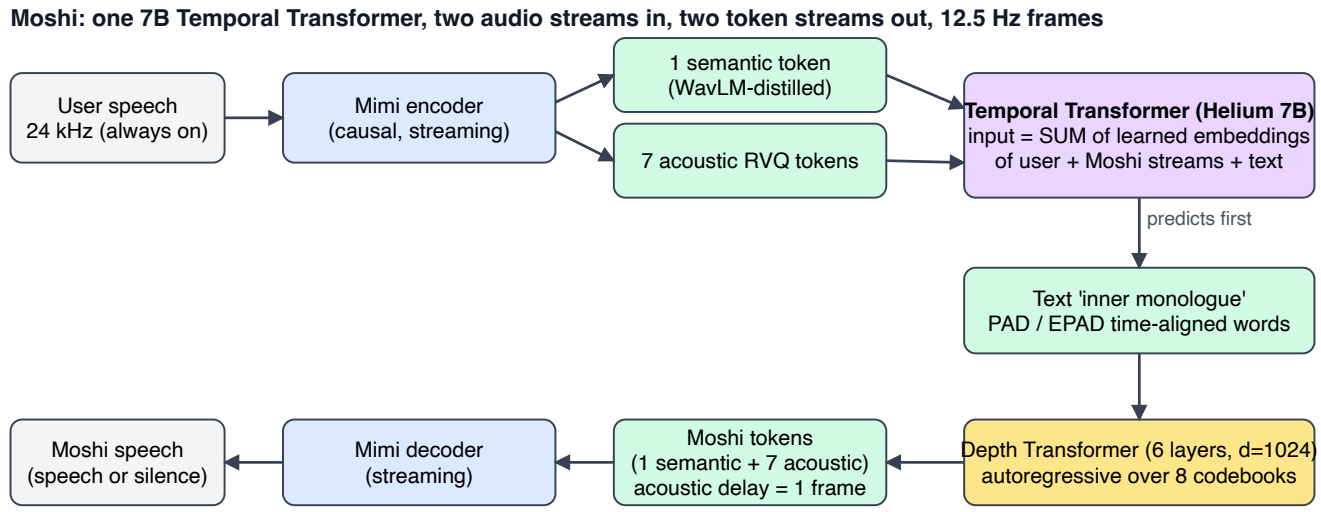


Fig. 1. Moshi data flow, redrawn from the paper's description.

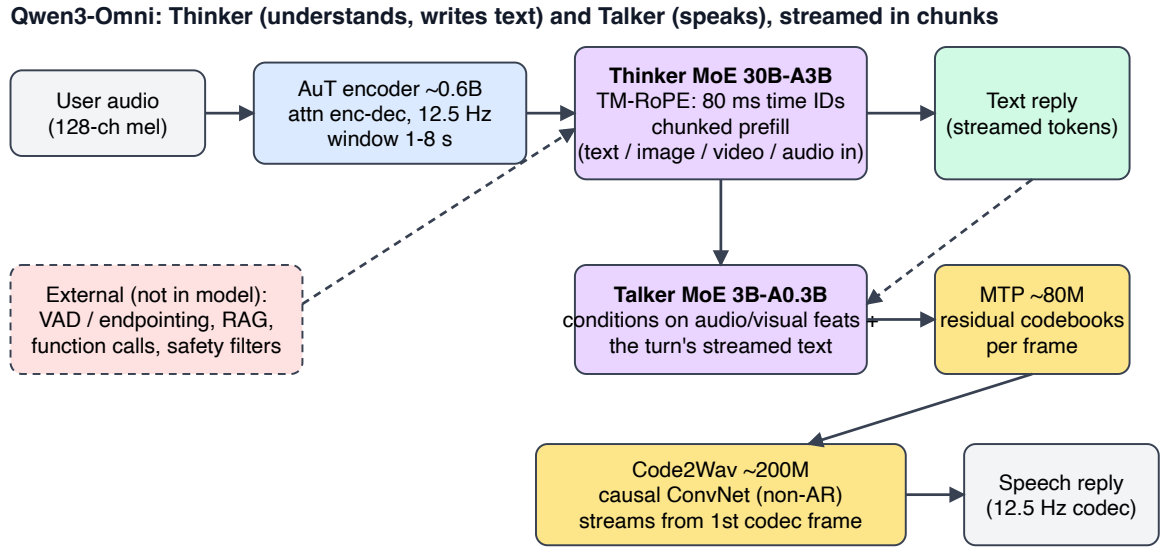


Fig. 2. Qwen3-Omni data flow. The dashed red box is my inference about where endpointing sits.

1.3 Research question: what blocks low-latency, natural, real-time Speech-LLM conversation?

Reading the two designs against each other, I see five limitations. Each is tied to a number or design choice in the reports.

1. **The latency floor is set by the representation, not the LLM.** Moshi's 160 ms is two 80 ms frames; going lower means a higher frame rate, which multiplies sequence length. A 12.5 Hz × 8-codebook stream is 100 audio tokens per second against roughly 3–4 text tokens per second for the same content, so the backbone spends ~25–30× more steps per second of content than a text LLM. The Depth Transformer and MTP module exist to hide this cost.
2. **Full-duplex and strong reasoning have not been combined.** Moshi gets ~200 ms and natural overlap, but it is a 7B model with a 4-minute context trained mostly on English. Qwen3-Omni has a far stronger Thinker but a turn-based interface, so the system must decide *when the user has finished* before it can commit to answering. Any silence timeout used for that decision adds directly to response time and cannot distinguish a thinking pause from a finished turn.
3. **Latency is not constant under load.** Qwen3-Omni's own table shows first-packet latency rising from 234 ms to 1172 ms between 1 and 6 concurrent streams, driven by Thinker and Talker time-to-first-token (88 → 673 ms and 57 → 376 ms). Real-time behaviour therefore depends on serving, not only architecture.
4. **Speech tokens carry less reasoning ability than text.** Moshi needed the inner-monologue text stream because predicting audio tokens directly gave poorer linguistic quality (their ablation ranks it among the most critical design choices). Qwen3-Omni likewise routes everything through a text-producing Thinker (*inference:* plausibly for the same reason; the report does not say). Either way, the system speaks only as fast as text plans ahead, and prosody that depends on *future* content (emphasis, question intonation) must be guessed.
5. **Coverage of languages that matter here.** AuT's pre-training mix is 80% Chinese/English and 10% other-language ASR; the 19 supported speech-input languages contain no Hindi or Tamil (Urdu is the only Indic-region language). Moshi is English-focused. My Task 2 results show the practical consequence: with a small multilingual stack, Hindi is limited by ASR errors and Tamil by the language model's comprehension, and both feed confident wrong answers.

1.4 Proposed improvement: Speculative Think-while-Listening (SpecTL)

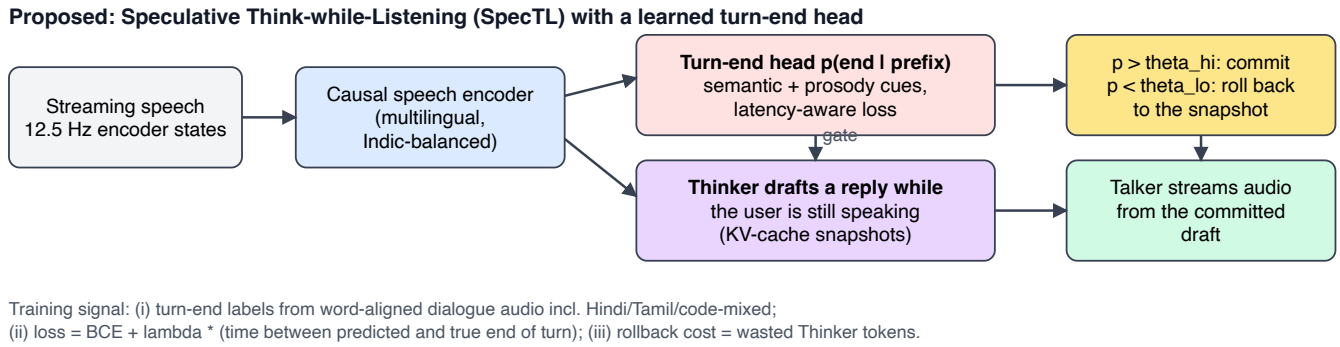


Fig. 3. Proposed gate between a streaming encoder and a turn-based Thinker–Talker.

Idea. Keep a strong turn-based Thinker–Talker (limitations 2 and 4) but remove the silence-timeout wait. A light head on the 12.5 Hz encoder states predicts the probability that the user's turn is ending, using semantic and prosodic cues. While the user is still speaking the Thinker drafts a response behind a KV-cache snapshot. When the head's probability crosses an upper threshold the draft is committed and the Talker starts; if the user continues, the draft is discarded and the cache is rolled back. This is speculative decoding applied to turn-taking rather than to tokens.

Hypotheses. H1: a learned turn-end head with a latency-aware loss gives lower response latency than a fixed 500–700 ms silence timeout at the same false-cut rate. H2: the wasted Thinker compute from rollbacks is small enough (<15% of drafted tokens) to be paid for by the saved wait. H3: an encoder trained on Indic data with the same head transfers poorly from English turn-end cues (Indic languages differ in final particles and pause structure), so an Indic-balanced head is needed.

Experiment design. (a) Data: two-channel dialogue with word-aligned turn ends; English (Fisher-like) plus Hindi/Tamil and code-mixed conversational audio. (b) Baselines: fixed-silence VAD, Moshi-style implicit turn-taking.

(c) Metrics: response onset minus true turn end (ms), false-cut rate, wasted draft tokens, and answer quality (judge score) so speed is not bought with worse answers. (d) Ablations: head inputs (semantic only, prosody only, both), loss weight λ, threshold pair, Indic vs English-only training. **Status: this is a proposal; I have not run it.**

Task 2 — Indic speech → text-response system and evaluation

> **What was built.** A reproducible pipeline: speech → ASR → LLM → English text response, evaluated on 120 FLEURS test utterances (60 Hindi, 60 Tamil; 23.7 min of audio) under two acoustic conditions (clean; white noise at 5 dB SNR), with my own evaluation code (WER/CER, response correctness, hallucination, latency). All models are open-source and run locally on one Apple M2 Pro (16 GB, MPS). The "response" task is *say in English what the speaker said*: the only task for which a reference answer exists for every clip (FLEURS clips are aligned to FLORES English text), which makes correctness and hallucination measurable without a human judge.

2.1 Setup and metric definitions

- **Data.** FLEURS test split, Hindi and Tamil, one clip per sentence id drawn with a fixed seed (0). English reference = the FLORES-aligned English sentence for the same id. **Noisy condition** = clean audio + white Gaussian noise at 5 dB SNR, seeded per utterance.
- **Baseline models.** ASR: openai/whisper-small (language forced to the known language, greedy). LLM: Qwen2.5-3B-Instruct (bf16, greedy), prompted with the transcript to reply with a faithful English translation and not to add information.
- **ASR metrics.** WER and CER against the FLEURS raw transcript, both sides NFC-normalised, lower-cased, punctuation stripped (incl. danda). Corpus-level; 95% CI by bootstrap over utterances.
- **Response correctness.** Cosine similarity (all-MiniLM-L6-v2) between the response and the English reference; **correct** if cos ≥ 0.6. I also report chrF.
- **Hallucination.** A response is flagged if any of: (H1) it states a number absent from the reference; (H2) semantic drift, cos < 0.4 (content unrelated to what was said); (H3) it is more than 2× longer than the reference + 5 words (content added beyond a translation).
- **Latency.** Wall-clock per stage with MPS synchronisation, batch size 1; real-time factor RTF = processing time ÷ audio duration (RTF < 1 is faster than real time). Measured in a dedicated benchmark (see 2.5).

> **Honest limits of the response metrics.** Correctness and hallucination are *automatic proxies*, not human judgements. The 0.6 / 0.4 thresholds were set by me after reading gold-transcript outputs across the cosine range (<0.4 unrelated, ≈0.45 half wrong, 0.6 mostly faithful, ≥0.73 faithful); they were not calibrated against human labels. Treat absolute percentages as indicative and the *comparisons between systems* as the result. A human-labelled audit of ~100 responses is the obvious next step. The embedding model also scores Tamil/Hindi-flavoured transliterations of names loosely, which depresses scores somewhat equally for all systems.

2.2 Baseline results

Clean audio:

System	WER %	CER %	Correct % [95% CI]	Hallucinated % [95% CI]	chrF
A0 whisper-small → Qwen2.5-3B (baseline)	69.3	27.6	9 [4–14]	74 [67–82]	24.7
B whisper-small → NLLB → Qwen	69.3	27.6	37 [28–45]	39 [30–48]	34.6
B′ whisper-small → NLLB only (control, no LLM)	69.3	27.6	36 [28–45]	37 [28–45]	34.8
A large-v3-turbo → Qwen	46.8	15.8	25 [17–33]	56 [47–64]	30.9
A+B large-v3-turbo → NLLB → Qwen	46.8	15.8	57 [48–65]	28 [21–37]	42.7
A+B′ large-v3-turbo → NLLB only (control)	46.8	15.8	57 [49–66]	28 [21–37]	43.3

C IndicConformer → Qwen	19.4	8.8	39 [31–48]	51 [42–60]	34.7
C+B IndicConformer → NLLB → Qwen	19.4	8.8	91 [85–96]	8 [4–13]	54.0
C+B′ IndicConformer → NLLB only (control)	19.4	8.8	88 [82–94]	8 [3–12]	54.8

5 dB noise:

System	WER %	CER %	Correct % [95% CI]	Hallucinated % [95% CI]	chrF
A0 whisper-small → Qwen2.5-3B (baseline)	97.7	56.9	4 [1–8]	94 [89–98]	17.6
B whisper-small → NLLB → Qwen	97.7	56.9	7 [2–12]	81 [73–88]	19.4
B′ whisper-small → NLLB only (control, no LLM)	97.7	56.9	4 [1–8]	81 [73–88]	19.0
A large-v3-turbo → Qwen	72.0	36.3	8 [3–13]	78 [71–85]	24.3
A+B large-v3-turbo → NLLB → Qwen	72.0	36.3	24 [17–32]	57 [48–66]	28.9
A+B′ large-v3-turbo → NLLB only (control)	72.0	36.3	23 [16–31]	55 [47–64]	28.5
C IndicConformer → Qwen	28.1	14.4	33 [25–42]	54 [45–63]	33.2
C+B IndicConformer → NLLB → Qwen	28.1	14.4	75 [67–82]	17 [11–23]	47.7
C+B′ IndicConformer → NLLB only (control)	28.1	14.4	77 [68–84]	16 [10–22]	49.5

Rows B′ and A+B′ are controls explained in 2.4. WER/CER do not depend on the LLM stage, so rows sharing an ASR share WER/CER.

Performance across conditions

ASR + LLM	Audio	Language	WER %	CER %	Correct %	Halluc. %
whisper-small → Qwen2.5-3B (baseline)	clean	Hindi	66.9	33.2	17	65
whisper-small → Qwen2.5-3B (baseline)	clean	Tamil	72.7	22.9	2	83
whisper-small → Qwen2.5-3B (baseline)	noisy	Hindi	97.6	69.5	7	90
whisper-small → Qwen2.5-3B (baseline)	noisy	Tamil	97.8	46.4	2	98
large-v3-turbo → Qwen	clean	Hindi	32.7	13.6	48	23
large-v3-turbo → Qwen	clean	Tamil	67.6	17.7	2	88
large-v3-turbo → Qwen	noisy	Hindi	58.3	30.7	15	60
large-v3-turbo → Qwen	noisy	Tamil	92.4	40.9	2	97
IndicConformer → Qwen	clean	Hindi	12.9	4.8	70	18
IndicConformer → Qwen	clean	Tamil	29.1	12.0	8	83
IndicConformer → Qwen	noisy	Hindi	21.1	11.0	62	23
IndicConformer → Qwen	noisy	Tamil	38.6	17.3	5	85

Baseline whisper-small is weak on Indic speech: 69% WER on clean audio and 98% at 5 dB, i.e. nearly every word is wrong under noise. Tamil has a much lower CER than WER (23% vs 73%) because it is agglutinative: one wrong suffix makes the whole word wrong, so WER overstates how much of the text is wrong. Hindi is hurt more by noise (CER 33% → 70%).

ASR systems compared

WER / CER (%), corpus level, on the 120 translation-set clips (60 per language). IndicConformer was run with both of its decoders; RNNT was fixed in advance for the downstream stages.

ASR system	Clean: WER / CER %		5 dB noise: WER / CER %	
	Hindi	Tamil	Hindi	Tamil
whisper-small (greedy)	66.9 / 33.2	72.7 / 22.9	97.6 / 69.5	97.8 / 46.4
whisper-large-v3-turbo (greedy)	32.7 / 13.6	67.6 / 17.7	58.3 / 30.7	92.4 / 40.9
IndicConformer-600M, CTC decoding	12.4 / 4.5	31.4 / 12.8	20.2 / 8.9	40.0 / 15.7
IndicConformer-600M, RNNT decoding	12.9 / 4.8	29.1 / 12.0	21.1 / 11.0	38.6 / 17.3

2.3 Bottleneck analysis

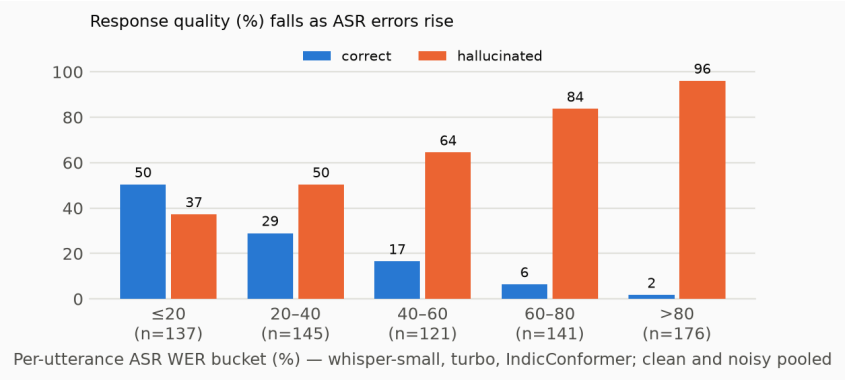


Fig. 4. Response quality against per-utterance ASR error (whisper-small, turbo and IndicConformer; clean and noisy pooled). Correctness falls from 50% (WER ≤ 20%) to 2% (WER > 80%); hallucination rises from 37% to 96%.

ASR errors are the first measurable bottleneck: per-utterance WER and response similarity are negatively correlated (Spearman –0.57 for Hindi, –0.28 for Tamil in the baseline), and the LLM turns recognition errors into *fluent, confident, wrong* English (e.g. a mis-heard Hindi word turned “local company” into “data company”). But a second experiment shows ASR is not the whole story. Feeding **gold transcripts** to the same LLM removes ASR error entirely:

Input to response stage	Hindi		Tamil		All	
	Correct %	Halluc. %	Correct %	Halluc. %	Correct %	Halluc. %
Gold transcript → Qwen	78	8	17	65	48	37
Gold transcript → NLLB only	97	3	88	8	92	6
Gold transcript → NLLB → Qwen	98	3	88	8	93	6

With perfect Hindi text Qwen2.5-3B is mostly adequate (78% correct), so for Hindi the bottleneck is ASR. With perfect **Tamil** text it gets only 17% correct and hallucinates 65%: for Tamil the bottleneck is the LLM's comprehension. This is why upgrading the ASR helps Hindi a lot (WER 67 → 33%, correct 17 → 48%) but does nothing for Tamil responses (2% → 2%), even though Tamil WER also improves slightly (73 → 68%).

2.4 Research extension: implementation and ablation

Bottleneck (measured on the whisper-small baseline, see 2.3): ASR for Hindi, and the 3B LLM’s weak Tamil comprehension for Tamil. **Improvements tested** (a factorial plus controls, all on the same 120 clips):

- **A — stronger ASR:** whisper-large-v3-turbo instead of whisper-small.
- **C — Indic-specialised ASR:** AI4Bharat IndicConformer-600M (hybrid CTC/RNNT Conformer trained on Indian languages; MIT licence, gated download), RNNT decoding chosen before looking at results; runs on CPU through ONNX.
- **B — translation grounding:** a dedicated Indic→English MT model (NLLB-200-distilled-600M) translates the transcript; the LLM receives the transcript *and* the draft translation and is told to use the draft as a guide.

- **B′ — control (no LLM):** the NLLB translation is returned directly. This tests whether the LLM adds anything beyond MT.

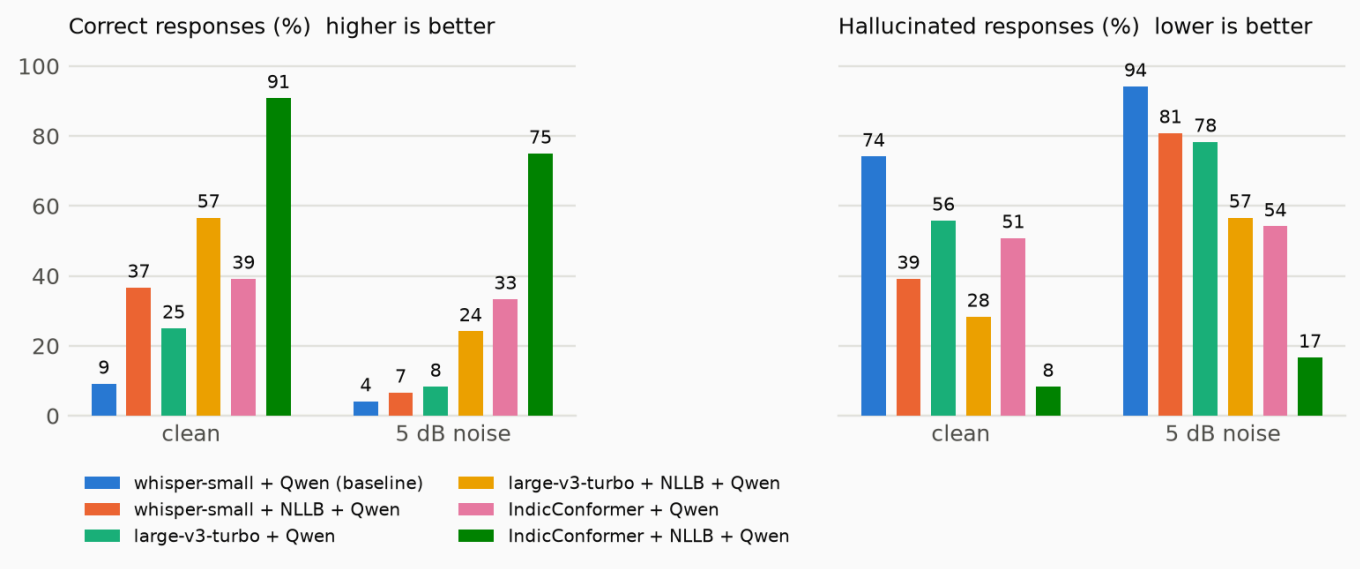


Fig. 5. Correct and hallucinated responses by system (all 120 clips per condition). Bars carry their values, so the chart does not depend on colour alone.

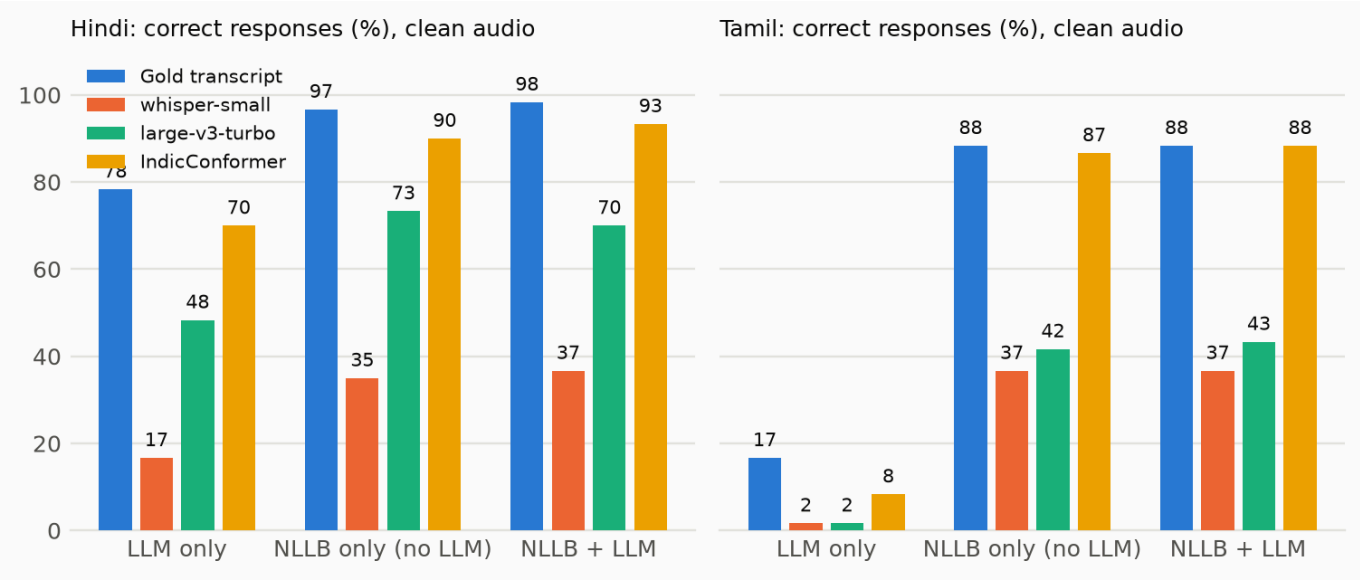


Fig. 6. Ablation by language on clean audio: which transcript the response stage sees (gold, whisper-small, turbo, IndicConformer) × which response stage runs.

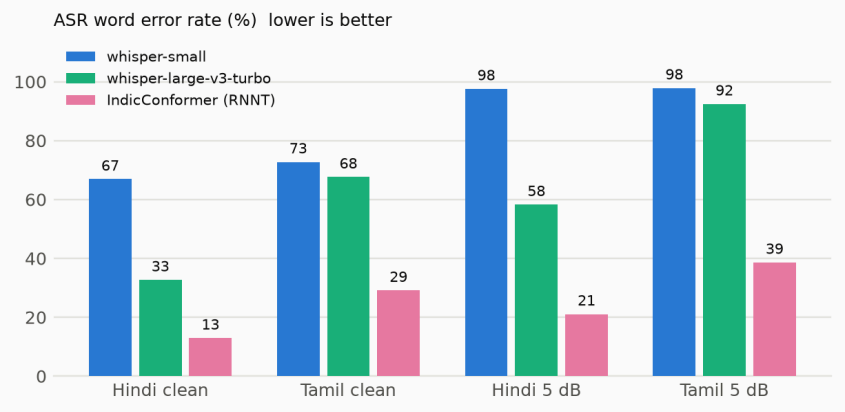


Fig. 7. ASR word error rate by language and condition. Turbo roughly halves Hindi WER but barely moves Tamil; the Indic-specialised IndicConformer cuts WER substantially in both languages and under noise.

Findings.

1. **The best system is IndicConformer + NLLB grounding, and it is far better than the baseline.** On clean audio, correct responses rise from 9% to 91% and hallucination falls from 74% to 8%. At 5 dB noise, correct rises from 4% to 75%. The bootstrap intervals in 2.2 do not overlap between the baseline and the best system.
2. **The ASR model is the biggest single lever, and a specialised model beats general ones.** Clean-audio WER falls from 69% (whisper-small) to 47% (whisper-large-v3-turbo) to 19% (IndicConformer); Hindi 67 → 33 → 13%, Tamil 73 → 68 → 29%. Turbo barely helps Tamil; IndicConformer does. It is also by far the most noise-robust: at 5 dB white noise its WER is 28% against 72% for turbo and 98% for whisper-small.
3. **Grounding and ASR are complementary, and grounding is what rescues Tamil.** With IndicConformer text, plain Qwen reaches only 8% correct on Tamil but 70% on Hindi; adding the NLLB draft lifts them to 88% and 93%. With the weaker whisper-small transcripts the same grounding takes Tamil from 2% to 37%.
4. **Negative finding: on this task the LLM adds essentially nothing over the MT model.** Returning NLLB's translation directly (no LLM) scores 88% correct with IndicConformer against 91% for NLLB + Qwen, a difference well inside the noise at n = 120, and the same pattern held for whisper-small and turbo. The gain from "grounding" is therefore the dedicated MT model replacing the LLM's weak Indic reading, and I do not present it as an LLM improvement. Whether the LLM helps when the task needs reasoning is tested in 2.7, where it is essential.
5. **This metric is near its ceiling.** Feeding *gold* transcripts through NLLB gives 92% correct, so the best system (91%) is within a few points of the best this proxy can show, and differences among the top systems are not resolvable here. That is why I added an exact-answer task (2.7).
6. **Tamil is still the harder language.** IndicConformer's Tamil WER (29%) is more than twice its Hindi WER (13%), partly because Tamil is agglutinative (CER 12% vs WER); on the comprehension task (2.7) Tamil stays close to chance even with gold text.

2.5 Latency

Measured in a separate benchmark on a fixed 24-clip clean subset (12 Hindi + 12 Tamil, 300 s of audio), one model in memory at a time. **The laptop was not idle:** swap usage at the start was total = 9216.00M used = 8256.69M free = 959.31M (encrypted), i.e. memory pressure from other applications. Absolute times are therefore probably pessimistic, and a quiet machine should be faster; the comparison *between* pipelines is the reliable part, because every stage ran under the same conditions. Stage times are additive, so each pipeline's latency is the sum of its stages.

Pipeline	Mean latency per utterance (s)	RTF
whisper-small → Qwen (baseline)	3.41	0.27
large-v3-turbo → Qwen	2.94	0.24
large-v3-turbo → NLLB only	2.16	0.17
large-v3-turbo → NLLB → Qwen	4.22	0.34

Stage	Mean seconds per utterance
ASR: whisper-small	1.67
ASR: whisper-large-v3-turbo	1.20
NLLB-600M translation	0.96
Qwen2.5-3B, transcript only	1.74
Qwen2.5-3B, transcript + NLLB draft	2.05

All pipelines run faster than real time (RTF < 1) on this laptop, and turbo is *faster* than whisper-small despite being larger because it decodes with a 4-layer decoder. Grounding costs about one extra NLLB call. IndicConformer runs on CPU through ONNX (no MPS), so it is not in this GPU benchmark; in the full evaluation run its ASR RTF was 0.05 (clean), i.e. about 18× faster than real time, but that figure was taken under memory pressure and with the

CPU shared, so treat it as indicative. These are batch (non-streaming) offline numbers: the whole utterance must be heard before any processing starts, so end-to-end *response latency after the user stops speaking* is the figure above, not a streaming first-token latency. The timing columns in `results/summary.csv` come from the full evaluation runs, during which memory pressure varied a lot (one run was visibly distorted by swapping), so use this benchmark rather than those columns.

2.7 Listening comprehension: an exact-answer response task (Belebele-Fleurs)

The translation task in 2.1–2.6 relies on proxy metrics, and the LLM turned out to add little there. To get a response task with an **exact** answer and one where the LLM has to reason, I added a spoken-passage multiple-choice test, built on *Belebele-Fleurs* (WueNLP/belebele-fleurs, CC-BY-SA): a FLORES passage is **spoken** (one FLEURS recording per sentence), and a **written** four-option question about it is asked. Pipeline: passage audio → ASR → (optionally NLLB translation to English) → Qwen2.5-3B, which picks the option letter. The letter is chosen as the argmax of the model's next-token probabilities over A/B/C/D (no sampling), so scoring is deterministic and **accuracy is exact**; chance is 25%.

- **Sample.** The first 100 questions per language in the dataset's own order (Hindi: first shard; Tamil: first two shards), i.e. 200 questions over 424 distinct sentence clips (92.7 min of audio). They are not randomly sampled, and the sample is small (95% Wilson intervals are about ±9 points per language).
- **Conditions.** Four passage sources: gold text (the ceiling), whisper-small, whisper-large-v3-turbo and IndicConformer-600M (RNNT, chosen in advance). Two prompt modes: passage, question and options in the original language (“native”), or all translated to English by NLLB-200-600M.
- **Confidently wrong.** I report the share of questions where the model picked a wrong option with probability ≥ 0.7, the multiple-choice analogue of a confident hallucination.

Passage source	Prompt	Hindi acc. % [95% CI]	Tamil acc. % [95% CI]	All (n=200)	Passage WER % (hi / ta)
Gold transcript (upper bound)	native-language prompt	59 [49–68]	41 [32–51]	**50 [43–57]**	
Gold transcript (upper bound)	NLLB-translated to English	65 [55–74]	47 [38–57]	**56 [49–63]**	
whisper-small	native-language prompt	39 [30–49]	35 [26–45]	**37 [31–44]**	68 / 70
whisper-small	NLLB-translated to English	45 [36–55]	36 [27–46]	**40 [34–47]**	68 / 70
whisper-large-v3-turbo	native-language prompt	55 [45–64]	37 [28–47]	**46 [39–53]**	30 / 61
whisper-large-v3-turbo	NLLB-translated to English	63 [53–72]	43 [34–53]	**53 [46–60]**	30 / 61
IndicConformer (RNNT)	native-language prompt	57 [47–66]	36 [27–46]	**46 [40–53]**	9 / 23
IndicConformer (RNNT)	NLLB-translated to English	60 [50–69]	44 [35–54]	**52 [45–59]**	9 / 23

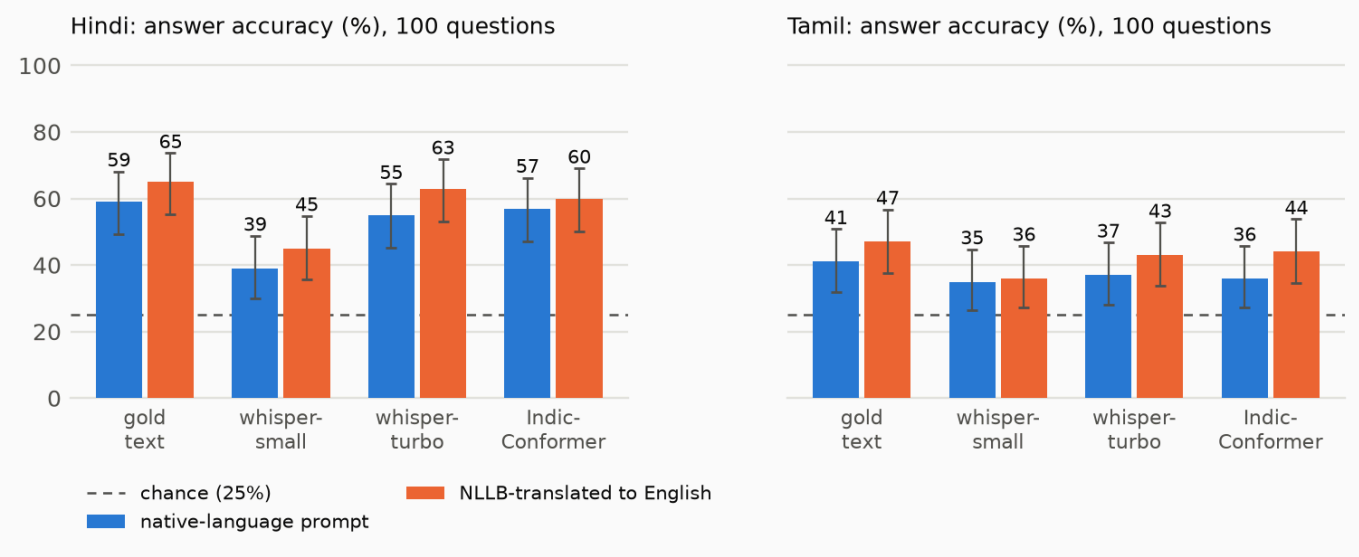


Fig. 8. Answer accuracy by passage source and prompt mode. Error bars are 95% Wilson intervals; the dashed line is chance (25%). With 100 questions per language the intervals are wide.

Findings.

1. **Better ASR improves comprehension, significantly.** With English prompts, IndicConformer beats whisper-small by +11.5 pts [+4.0, +19.0] on all 200 questions (paired bootstrap; Hindi +15.0 pts [+4.0, +26.0], Tamil +8.0 pts [-3.0, +19.0]), and turbo beats whisper-small by +12.5 pts [+6.5, +18.5]. The Tamil interval includes zero, so the effect is established for Hindi and pooled data, not for Tamil alone.
2. **The ceiling is the LLM, not the ASR.** Even with *gold* text Qwen2.5-3B answers only 56% correctly with English prompts, and for Tamil 41–47% against 25% chance. IndicConformer is within +4.0 pts [-1.0, +9.0] of that ceiling (interval includes zero), so there is little left for ASR to gain on this model. This matches the translation experiment, where Tamil was also capped by the LLM.
3. **Lower WER does not always buy more accuracy.** IndicConformer has far lower passage WER than turbo (Hindi 9% vs 30%, Tamil 23% vs 61%), but the two reach similar accuracy (52% vs 53%, English prompts): once WER is below roughly 30%, the LLM's comprehension is the binding constraint.
4. **Translating to English helps a little.** English prompts score higher than native prompts in every pairing (+6.0 pts [-2.5, +14.5] with gold text, +5.5 pts [-3.0, +14.5] with IndicConformer), but these intervals include zero at n = 200, so I treat it as suggestive rather than established.
5. **Models are confidently wrong.** Depending on source and mode, 39–56% of all questions are answered wrongly with probability ≥ 0.7; this stays high even with gold text, so it reflects model overconfidence rather than ASR damage alone.

Checks: predicted letters are spread over all four options (D is chosen somewhat less often, about 14–18% of the time against 20% of gold answers), so accuracy is not an artefact of always answering one letter. Only the transcription stage, not the question, is spoken; a fully spoken question-and-answer test, other LLMs, and larger samples are left for future work.

2.8 Threats to validity

- **Small sample, one seed.** 120 utterances (60 per language); intervals are wide for per-language cells (n = 60).
- **Proxy metrics** (see 2.1). No human audit.
- **Read speech.** FLEURS is read, clean-studio-ish speech; spontaneous Indic speech will be harder. Noise is synthetic white noise, harsher and less structured than real babble or street noise.
- **Task choice.** Translation is a narrow notion of "response"; it was chosen so correctness is measurable. The LLM-adds-little finding is specific to this task.
- **Thresholds were fixed after viewing outputs** (disclosed above); I re-scored correctness at 0.5 and 0.7 (`correct_0.5`, `correct_0.7` in `results/per_utterance.csv`). Across the nine systems, the coarse

ordering is stable at 0.5, 0.6 and 0.7: on clean audio the four best (IndicConformer + NLLB + Qwen, its no-LLM control, turbo + NLLB + Qwen, its control) and the two worst (turbo + Qwen, baseline) keep their places, while the middle ranks (plain IndicConformer + Qwen, and the whisper-small + NLLB variants) swap. On noisy audio the three IndicConformer systems are always the top three, followed by the turbo systems; the three whisper-small systems all score ≤ 11% and trade places, and IndicConformer + NLLB + Qwen and its no-LLM control swap first and second, so those two should be read as indistinguishable.

- **Language is forced** for Whisper (known-language setting); automatic language ID was not evaluated.

Appendix: example outputs

The first two utterances of each language in manifest order (no selection by outcome).

hi_1828 · reference (English): A search of the Internet for 'Hostile environment course' will probably provide the address of a local company.
Reference transcript: इंटरनेट पर यह खोज 'शत्रुतापूर्ण पर्यावरण पाठ्यक्रम' के लिए अक्सर आपको एक स्थानीय कंपनी का पता प्रदान करेगी।
Baseline ASR: अंटरनेट पर ये खोज सत्रुता पुर्न पर्यावरं पाटक्रम के लिये अकसर आपको एक इस धानने कमपनी का पता प्रदान करेगी
Baseline response (cos 0.18): On the internet, it will sometimes show you the name of a data company to resolve this dispute fully.
Turbo ASR: इंटरनेट पर ये खोज सत्रुतापुर्ण परियावरंग पाठिकरम के लिए अक्सर आपको एक इस्थाने कमपनी का पता प्रदान करेगी।
Turbo+NLLB+Qwen response (cos 0.30): This search on the Internet will often give you the address of a stationary company for the saturated Pattikaram readership.

hi_1815 · reference (English): The fission bomb works on the principle that it takes energy to put together a nucleus with many protons and neutrons.
Reference transcript: विखंडन बम इस सिद्धांत पर काम करता है कि कई प्रोटॉन और न्यूट्रॉन वाले एक नाभिक को एक साथ रखने के लिए ऊर्जा लगती है।
Baseline ASR: भिख्छन्दन भम इस सिद्दान्त पर काम करता है कि कई प्रोटोन और नूट्रोन वाले एक नाभिक को एक साथ रकने के लिए उर्जा लकती है.
Baseline response (cos 0.60): Poisoning Bhamb works on the principle that energy is available to fuse several proton and neutron-rich nuclei into one.
Turbo ASR: विखंडन बम इस सिध्धान्त पर काम करता है कि कई प्रोटॉन और न्यूट्रॉन वाले एक नाभिक को एक साथ रखने के लिए उर्जा लगती है।
Turbo+NLLB+Qwen response (cos 0.99): The fission bomb works on the principle that it requires energy to keep a nucleus with many protons and neutrons together.

ta_1878 · reference (English): The central authority of the church had been in Rome for over a thousand years and this concentration of power and money led many to question whether this tenet was being met.
Reference transcript: சர்ச்களின் மத்திய அதிகாரம் கடந்த ஆயிரம் ஆண்டுகளுக்கு மேலாக ரோமில் இருந்து வருகிறது மற்றும் இந்த அதிகாரம் மற்றும் செல்வத்தின் ஒரே இடத்திலான செறிவு, பல பேரை அடிப்படைக் கோட்பாடுகள் நிறைவேற்றப் படுகின்றனவா என்ற கேள்விக்கு இட்டுச் செல்கிறது.
Baseline ASR: சர்ச்சுகளின் மத்திய அதிகாரம் கடந்த ஆயிரம் ஆண்டுகளுக்கு மேலாக ரோமிலிருந்து வருயிரத்து, மற்றும் இந்த அதிகாரம் மற்றும் செல்வத்தின் ஒரே இடத்தில் ஆண்டு செரிவு, பல பேரை அழிப்படைக் கோட்பாடுகள் நிறைவேற்ற படிக்குண்டுரணவா என்று கேள்வேக்கு இட்டுச் செல்கிறது.
Baseline response (cos 0.35): The Supreme Court of the State has moved from Rome to another location, and this court and its proceedings have caused many disputes to arise, as asked by Justice Kekkavu.
Turbo ASR: சர்ச்சுகளின் மத்திய அதிகாரம் கடந்த ஆயிரம் ஆண்டுகளுக்கு மேலாக ரோமில் இருந்து வரியிறது மற்றும் இந்த அதிகாரம் மற்றும் செல்வத்தின் ஒரை இடத்திலான செரிவு பழபேரை அடிப்படை கோட்பாடுகள் நிறைவேற்ற படிகின்றனமா என்று கேல்விக்கு இட்டுச் செல்கிறது
Turbo+NLLB+Qwen response (cos 0.71): The central authority of the churches has been drawing from Rome for over a thousand years, and this wealthy half-spouse of power is leading to Calvius questioning whether the basic principles are being fulfilled.

ta_1677 · reference (English): Today, the only insects that cannot fold back their wings are dragon flies and mayflies.
Reference transcript: இன்று, தங்கள் இறக்கைகளை மடிக்க முடியாத ஒரே பூச்சிகள் தும்பித் தட்டான் மற்றும் மேஃப்ளைகள் ஆகும்.
Baseline ASR: இன்று தங்கல் ஏறக்கைகளை மடிக்க முடியாத ஒரே பூச்சிகள் தும்பித் தட்டான் மட்டும் மேப்பளைகளாகும்
Baseline response (cos 0.01): Today's thunderstorms will be just showers.
Turbo ASR: இன்று தங்கள் இரக்கைகளை மடிக்க முடியாத ஒரே பூச்சிகள் தும்பி தட்டான் மட்டும் மேப்ப்பையிகள் ஆகும்
Turbo+NLLB+Qwen response (cos 0.61): The only insects that can't fold their ears today are the mapmakers.